

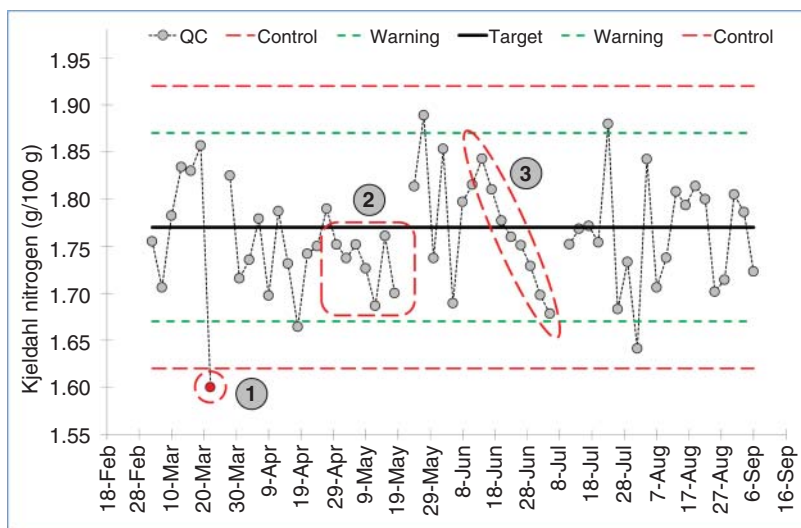


Figure 4.4 NITROGEN – examples of anomalies in a wheat flour control chart.

- ① One QC is outside one control limit.
- ② Seven successive QC are systematically above or below the target value, and as it is expected, they are randomly spread on both sides.
- ③ Eight successive QCs indicate a systematic upward or downward trend.

Moreover, if any other QC distribution suggests nonrandom behavior, the quality controller is free to take corrective action. The assumption underlying the control charts is that measurements made under statistical control should be distributed according to the Normal law. Control limits can be compared to outlier rejection limits, and it is assumed that any exceeding data corresponds to a malfunction in the measurement process. From a practical point of view, the control chart set up in a laboratory may be different from the industrial context. The easiest way is to use a CRM or an ERM, but their availability or their cost may make this strategy inapplicable. If the laboratory decides to use an IRM, the control chart set up consists of two phases.

4.4.1 First Phase Assessment of the Reference Value

1. Prepare homogenized material, in sufficient quantity to complete several tens of test portions. It is best to package each test sample in a sealed bag so that the composition of the material remains stable over time.
2. Analyze at least 30 test portions ($I \geq 30$) with the method to be controlled. When a QC is subsequently planned to involve an average of J replicates, all 30 analyses must be done under the same conditions.
3. Calculate the mean $\bar{Z}$, the standard deviation $\hat{\sigma}$, and the warning (WL) and control (CL) limits, as described in Eqs. (4.8) and (4.9). Then draw the control chart. The notation $\hat{\sigma}$ is used in this context rather than s because it is the traditional notation for control chart.

4.4.2 Second Phase Routine Use

1. Define a control frequency (one per day, for example) and analyze a different QC at each control, the answer is denoted X_i . If several test portions are analyzed for each QC, individual values are reported unless a result is expressed as the average of replicates. Section 8.4.3 gives more explanation.
2. When about 20 test portions remain, prepare another internal standard material, measure its contents to define its new reference value and its new standard deviation, and ensure continuity and traceability between old and new control charts.

The standard deviation $\hat{\sigma}$ and the mean T are obtained with the classic following formulas:

Target value

$$T = \bar{Z} = \frac{\sum_i Z_i}{I}$$

Standard deviation

$$\hat{\sigma} = \sqrt{\frac{\sum_i (Z_i - \bar{Z})^2}{I - 1}}$$

Control limits

$$LCL = T - 3\hat{\sigma} \quad UCL = T + 3\hat{\sigma} \quad (4.8)$$

Warning limits

$$LWL = T - 2\hat{\sigma} \quad UWL = T + 2\hat{\sigma} \quad (4.9)$$

If each QC corresponds to the average of J replicates, as usually recommended, WL and CL must be differently calculated. New formulas consider this improvement of the control:

Control limits

$$LCL = T - \frac{3\hat{\sigma}}{\sqrt{J}} \quad UCL = T + \frac{3\hat{\sigma}}{\sqrt{J}}$$

Warning limits

$$LWL = T - \frac{2\hat{\sigma}}{\sqrt{J}} \quad UWL = T + \frac{2\hat{\sigma}}{\sqrt{J}}$$

For instance, the ISO 7870-2 standard proposes a set of tables that allow the calculation of the limits where measures are expected to lie for Shewhart control charts [9]. Control charts with other distribution laws than the Normal law remain applicable when dealing with counts, such as microbiological methods. It is obvious that a control based on replicates is more efficient but also more expensive. It is reasonable when measurements are quick and simple, such as physical measurements, temperature, or weighting.

In Section 7.3, a method is proposed to directly derive control charts from the statistical tolerance intervals used to build the MAP. The different limits are obtained

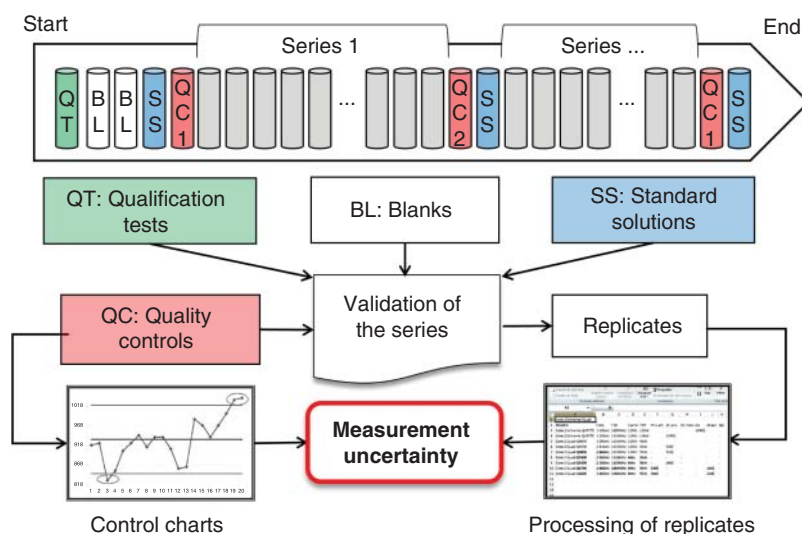


Figure 4.5 Possible location of different Qcontrol charts in a routine laboratory. Legend: QT, qualification test; BL, blank; SS, standard solution; QC1, QC2, quality controls.

using different values of the probabilities applied to define these intervals. Finally, control charts are a direct application of the principles of theoretical distribution laws. They are increasingly required by auditors of accreditation bodies, as well as participation in proficiency testing. Figure 4.5 shows a possible organization of different control charts for QC in a routine laboratory. The samples sent by the clients to the laboratory are grouped into series and surrounded by diverse types of controls, such as standard solutions, noted SE, blanks BL and, of course, RMs of known content, noted QC1, QC2, etc. Separate control charts can be established for each type of control. At the start of the production line, qualification tests (QT) are applied to the instruments. These tests can also include measurements on commercial reference solutions, as is recommended for many methods, for example, mass spectrometry. The fixed costs for such an organization are not negligible and substantiate the grouping of analyses as series. The remaining question is to define the frequency of the QC.

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